

AtliQ Hardware Ad hoc Analysis

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1. Provide the list of markets in which customer "Atliq Exclusive" operates its business in the APAC region.

```
select distinct market
from dim_customer
where customer = "Atliq Exclusive"
and region = "APAC";
```

market
India
Indonesia
Japan
Philippines
South Korea
Australia
Newzealand
Bangladesh

2. What is the percentage of unique product increase in 2021 vs. 2020? The final output contains these fields,

unique_products_2020
unique_products_2021
percentage_chg

```
with cte1 as
(
  select
    (select count(distinct p.product_code)
     from dim_product p
     join fact_sales_monthly s
     on p.product_code = s.product_code
     where s.fiscal_year = 2020) as unique_products_2020,

    (select count(distinct p.product_code)
     from dim_product p
     join fact_sales_monthly s
     on p.product_code = s.product_code
     where s.fiscal_year = 2021) as unique_products_2021)

select unique_products_2020, unique_products_2021,
round(((unique_products_2021 - unique_products_2020)/unique_products_2020)*100,2) as percentage_chg
from cte1;
```

unique_products_2020	unique_products_2021	percentage_chg
245	334	36.33

3. Provide a report with all the unique product counts for each segment and sort them in descending order of product counts. The final output contains 2 fields,

segment
product_count

```
select segment,  
count(distinct product_code) as product_count  
from dim_product  
group by segment  
order by product_count desc;
```

segment	product_count
Notebook	129
Accessories	116
Peripherals	84
Desktop	32
Storage	27
Networking	9

4. Follow-up: Which segment had the most increase in unique products in 2021 vs 2020? The final output contains these fields,

segment
product_count_2020
product_count_2021
difference

```
with products_2020 as
  (select p.segment,
   count(distinct p.product_code) as product_count_2020
   from dim_product p
   join fact_sales_monthly f
   on p.product_code = f.product_code
   where f.fiscal_year = 2020
   group by p.segment),
```

```
products_2021 as
  (select p.segment,
   count(distinct p.product_code) as product_count_2021
   from dim_product p
   join fact_sales_monthly f
   on p.product_code = f.product_code
   where f.fiscal_year = 2021
   group by p.segment)
```

```
select
x.segment,
x.product_count_2020, y.product_count_2021, (y.product_count_2021 - x.product_count_2020) as difference
from products_2020 x
join products_2021 y on x.segment = y.segment
order by difference desc;
```

segment	product_count_2020	product_count_2021	difference
Accessories	69	103	34
Notebook	92	108	16
Peripherals	59	75	16
Desktop	7	22	15
Storage	12	17	5
Networking	6	9	3

5. Get the products that have the highest and lowest manufacturing costs.

The final output should contain these fields,

product_code

product

manufacturing_cost

```
with high as( select
    p.product_code, p.product, m.manufacturing_cost
from fact_manufacturing_cost m
    join dim_product p on m.product_code = p.product_code
where m.manufacturing_cost =
    (select max(m.manufacturing_cost) from fact_manufacturing_cost m)),

low as( select
    p.product_code, p.product, m.manufacturing_cost
from fact_manufacturing_cost m
    join dim_product p on m.product_code = p.product_code
where m.manufacturing_cost =
    (select min(m.manufacturing_cost) from fact_manufacturing_cost m))

select product_code, product, manufacturing_cost
from high h union
select product_code, product, manufacturing_cost from low l;
```

product_code	product	manufacturing_cost
A6120110206	AQ HOME Allin1 Gen 2	240.5364
A2118150101	AQ Master wired x1 Ms	0.8920

6. Generate a report which contains the top 5 customers who received an average high pre_invoice_discount_pct for the fiscal year 2021 and in the Indian market. The final output contains these fields,
- customer_code
 - customer
 - average_discount_percentage

```
select pre.customer_code, c.customer,  
       avg(pre_invoice_discount_pct) as average_discount_percentage  
from fact_pre_invoice_deductions pre  
     join fact_sales_monthly f  
       on f.customer_code = pre.customer_code  
     join dim_customer c  
       on c.customer_code = pre.customer_code  
       where f.fiscal_year = 2021  
          and c.market = 'India'  
group by pre.customer_code, c.customer  
order by average_discount_percentage desc  
limit 5;
```

customer_code	customer	average_discount_percentage
90002009	Flipkart	0.29545000
90002003	Ezone	0.28195000
90002004	Vijay Sales	0.28115000
90002011	Atliq Exclusive	0.27245000
90002006	Viveks	0.27180000

7. Get the complete report of the Gross sales amount for the customer “**Atliq Exclusive**” for each month. This analysis helps to get an idea of low and high-performing months and take strategic decisions.

The final report contains these columns:

Month

Year

Gross sales Amount

```
select
monthname(f.date) as Month,
f.fiscal_year,
sum(f.sold_quantity*g.gross_price) as Gross_Sales_Amount,
c.customer
from fact_sales_monthly f
join fact_gross_price g
    on f.fiscal_year = g.fiscal_year
    and f.product_code = g.product_code
join dim_customer c
    on f.customer_code = c.customer_code
where c.customer = "Atliq Exclusive"
group by Month, f.fiscal_year, c.customer
order by f.fiscal_year, Month asc;
```

Month	fiscal_year	Gross_Sales_Amount
April	2020	395035.3535
August	2020	2786648.2601
December	2020	4830404.7285
February	2020	3996227.7661
January	2020	4740600.1605
July	2020	2551159.1584
June	2020	1695216.6008
March	2020	378770.9700
May	2020	783813.4238
November	2020	7522892.5608
October	2020	5135902.3467
September	2020	4496259.6724
April	2021	7311999.9547
August	2021	7178707.5902
December	2021	12944659.6509
February	2021	10129735.5675
January	2021	12399392.9788
July	2021	12092346.3245
June	2021	9824521.0110
March	2021	12144061.2501
May	2021	12150225.0139
November	2021	20464999.0997
October	2021	13218636.1966
September	2021	12353509.7938

8. In which quarter of 2020, got the maximum total_sold_quantity? The final output contains these fields sorted by the total_sold_quantity,

Quarter

total_sold_quantity

```
select
  case
    when month(date) IN (9,10,11) then "Q1"
    when month(date) IN (12,1,2) then "Q2"
    when month(date) IN (3,4,5) then "Q3"
    when month(date) IN (6,7,8) then "Q4"
  end as Quarter,
  sum(sold_quantity) as total_sold_qty
from fact_sales_monthly
where fiscal_year = 2020
group by Quarter
order by total_sold_qty;
```

Quarter	total_sold_qty
Q3	2075087
Q4	5042541
Q2	6649642
Q1	7005619

9. Which channel helped to bring more gross sales in the fiscal year 2021 and the percentage of contribution? The final output contains these fields,

channel
gross_sales_mln
percentage

```
with cte1 as(
select
c.channel,
round(sum(g.gross_price*f.sold_quantity)/1000000,2) as gross_sales_Mln
from dim_customer c
    join fact_sales_monthly f
    on c.customer_code = f.customer_code
    join fact_gross_price g
    on f.product_code = g.product_code
    and f.fiscal_year = g.fiscal_year
where f.fiscal_year = 2021
group by c.channel)

select *,
round(gross_sales_Mln*100/sum(gross_sales_Mln) over ( ),2) as percentage
from cte1;
```

channel	gross_sales_Mln	percentage
Direct	257.53	15.47
Retailer	1219.08	73.23
Distributor	188.03	11.30

10. Get the Top 3 products in each division that have a high total_sold_quantity in the fiscal_year 2021? The final output contains these fields,

division	product
product_code	total_sold_quantity
	rank_order

```
with cte1 as(
select
p.division, p.product_code, p.product,
sum(f.sold_quantity) as total_qty
from dim_product p
join fact_sales_monthly f
on p.product_code = f.product_code
group by p.product_code, p.product, p.division
order by total_qty desc),

cte2 as(
select *,
dense_rank() over (partition by division order by total_qty) as rank_order
from cte1)

select * from cte2 where rank_order <=3;
```

division	product_code	product	total_qty	rank_order
N & S	A7321160301	AQ Wi Power Dx3	281363	1
N & S	A7321160303	AQ Wi Power Dx3	286486	2
N & S	A7321160302	AQ Wi Power Dx3	291451	3
P & A	A3718150104	AQ LION x1	2018	1
P & A	A4021150405	AQ Mx NB	34361	2
P & A	A4021150403	AQ Mx NB	34683	3
PC	A5318110101	AQ Gamer 1	1515	1
PC	A5318110102	AQ Gamer 1	1562	2
PC	A6119110202	AQ HOME Allin1 Gen 2	2285	3